

Manhattan Manual Voltage Processor (MVP) Assembly and Wiring Information

DotCom Version

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NOTE: This document contains information originally published and/or obtained, directly or indirectly, from synthCube, Jason Coates, Muffwiggler and other sources. All credit goes to those folks and no infringement is intended. This document combines the information used and my experience constructing the module described obtained from synthCube.

I LOVE THIS MODULE aaaand I hate it. I love the simplicity and functions: three attenuverters, a dual/quad mixer, buffered and passive multiples. I hate that there's no Control PCB for the Dotcom version. But that makes life interesting.

The synthCube kits provided a small perf board on which to mount the LED drive opamp and resistors that are on the Eurorack version Control PCB. But there's no easy way to mount the PCB in the DotCom version. So you have a choice: simply don't do the LED section or build a secondary PCB to do the LEDs.

Or you could add the LEDs later.

BUT, if you intend to add the LEDs and Attenuverter Buffers later, make a template for the Secondary PCB and check that it aligns with the standoffs and LED locations. See the Secondary PCB section below.

Note: I came up with this approach about half way into writing this so some of the pictures may not quite match the text.

TIP: Photocopy the front and back of PCBs for reference. Blow it up as large as possible for easy viewing.

TIP: Always save any unused parts for other projects.

Assembling the MVP without the LEDs:

If you intend to add the LEDs and Attenuverter Buffers later, make a (paper) template or use the method later in this document below for the Secondary PCB and check that it aligns with the standoffs and LED locations. See the Secondary PCB Preparation section below.

Also, leave the wires that run from the Attenuverter pot wipers to the jacks long enough to reach back to the Main PCB so all you have to do is cut them in the middle later.

Overview:

- 1) Build the Main PCB
- 2) Install the jacks and pots into the Front Panel oriented per the diagram later in this document.
- 3) Do the Front Panel wiring per the diagram.
- 4) Install/stack both the 1" and ½" standoffs so that the Main PCB will clear the jacks. Or use the 1.25" standoffs listed below.
- 5) Hook the Main PCB upside down onto two left standoffs.
- 6) Use the "ABC" wiring information in this document for the connections. Leave about 1-2" slack in the wires to allow adding the Secondary PCB later.
- 7) Maybe plug the 5/16" holes where the LEDs would go or use the snap-in LEDs furnished with the kit but don't connect them.

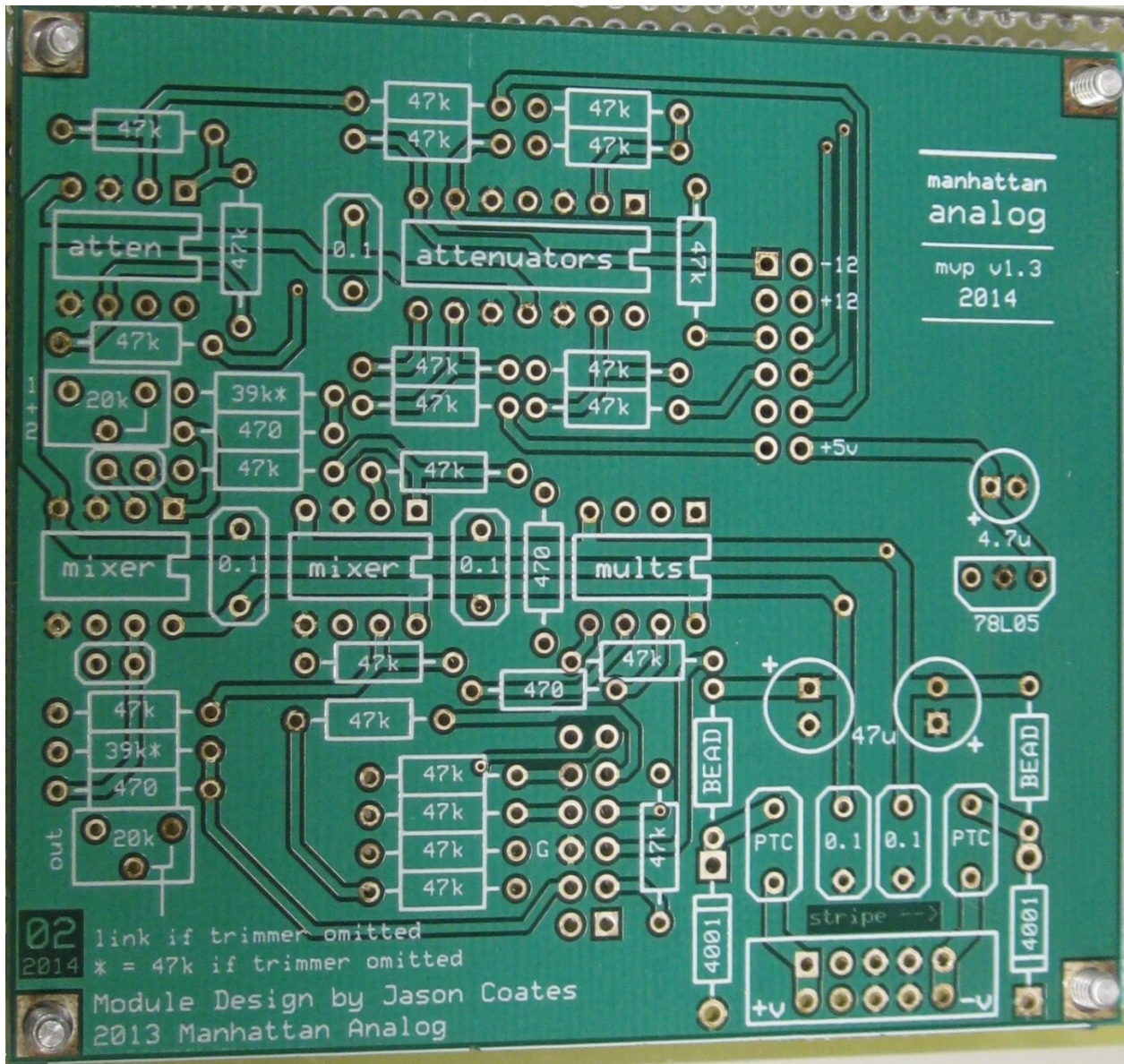
Just that simple, enjoy!

Main PCB Assembly:

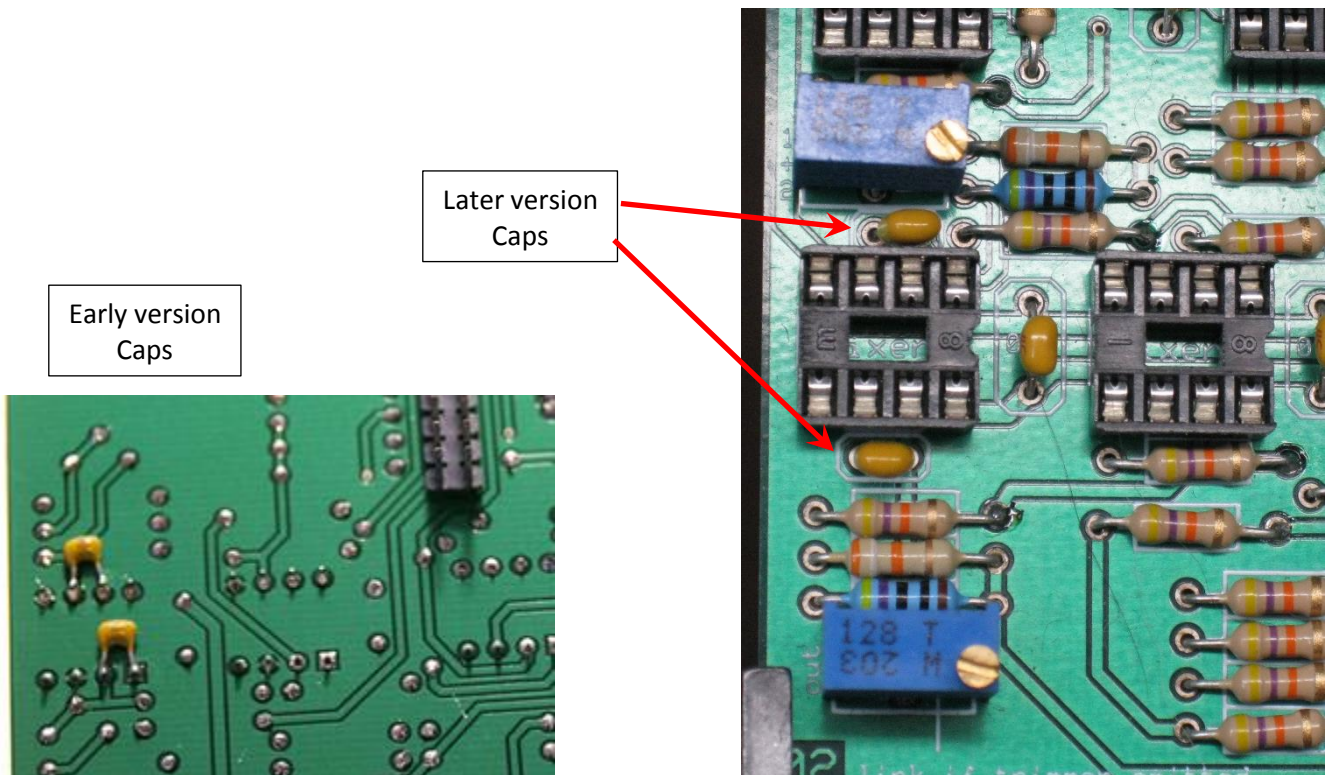
The Main PCB is a straight forward build and follows the published information reproduced here. These items are included in the synthCube kit.

Description	Quantity
470 Ohm resistor	4
39K resistor	2
47K resistor	23
22pF ceramic cap	2
0.1uF Ceramic cap	5
4.7uF electrolytic cap	1
10/22/47uF electrolytic cap	2
20K trimpot	2
LM7805	1
TL074	1
TL072	4
14 pin IC socket	1
8 pin IC socket	4
6 pin MTA header	1

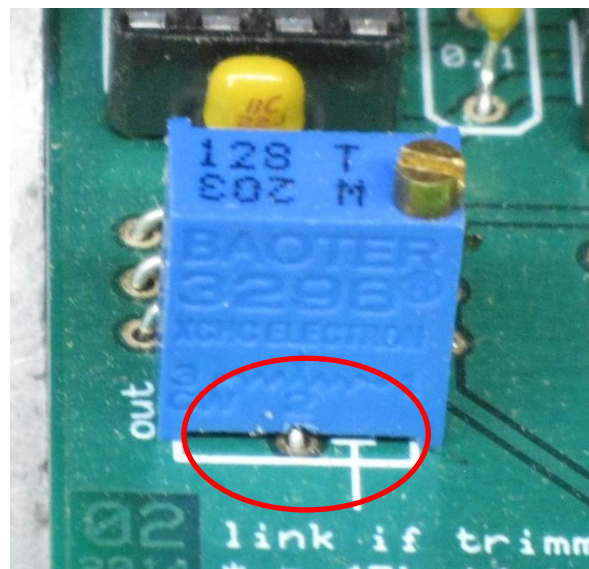
Jumpers may be used in place of the trimpots.
If so, replace the two 39K resistors with 47K resistors to produce unity gain.
(Not included in the kit.)



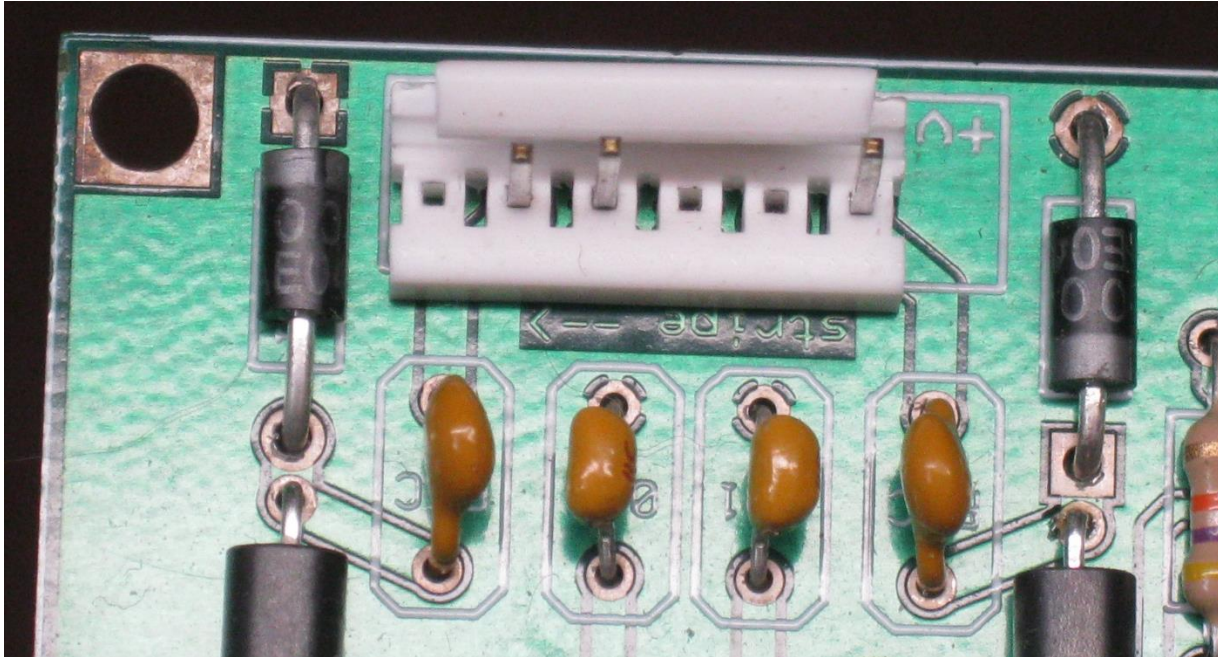
- The original build document describes the need to add two 22pF caps on the back. The kits I had were V1.3 and had provisions on the PCB for the caps but were not marked.



- The PCB was laid out for staggered pin trim pots but the kits had in-line pots. Cut a small notch in the plastic, bend the lead over and down. It will install just fine.



- The PCB is laid out for a 2x5 EuroRack power header and the kits furnished a MOTM PCB and DotCom header. Remove pins 2, 3 and 6 from the DotCom header and install as shown.



Note: Normally on non-DotCom modules that have a 5V regulator on them, I will wire in the +5V that DotCom power provides. However, as in this case, where the 5V is being used as a reference voltage, I use the on board regulator to provide the cleanest possible voltage source.

The Front Panel Preparation:

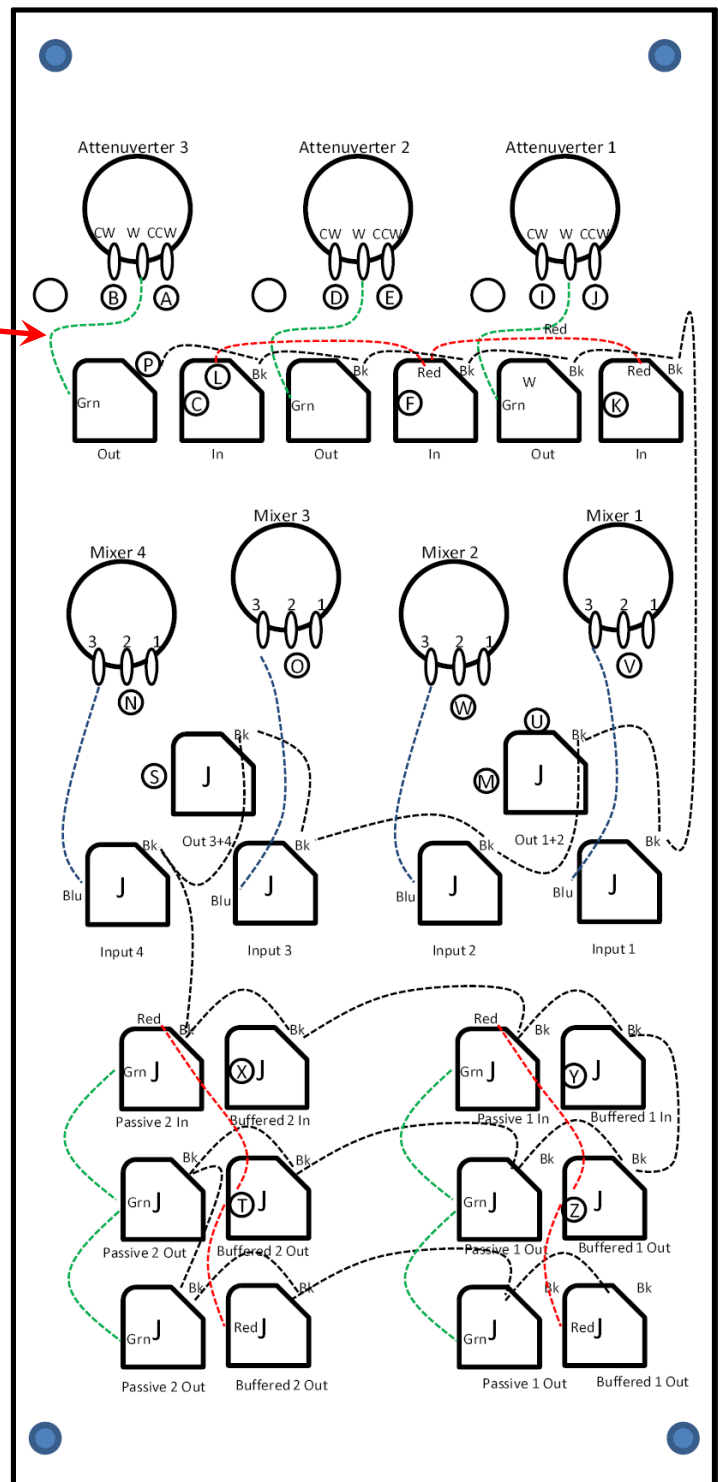
The Front Panel is fairly straight forward and follow the published information reproduced here except for the Attenuverter Outputs. On the original design, the Attenuverter output pot wiper was just fed to the jack and the LED driver. I've added a buffer which just adds three wires between the Secondary PCB and the jacks.

- 1) Remove the protective plastic and clean the panel if necessary.
- 2) Install the Pots and Jacks and wire as shown

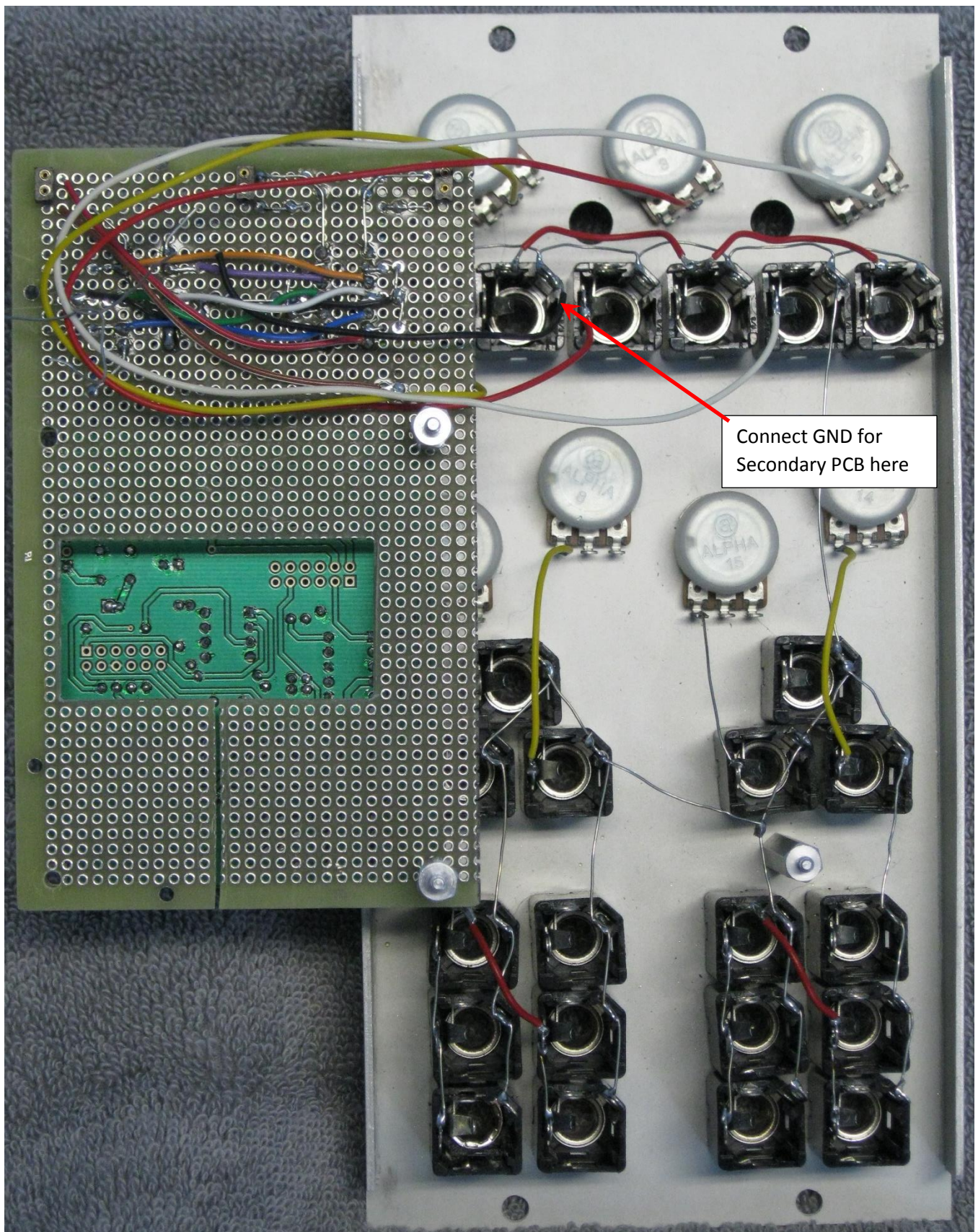
If you plan to add the LEDs later, make the wires from the Attenuverter pot's wiper to the jack long enough to reach the Main PCB. All you will need to do later is cut the wire in half and connect to the Secondary PCB.

See the picture below.

Image provided by
synthCube



MA MVP Rear View of Panel- dotcom/MU



Connect GND for
Secondary PCB here

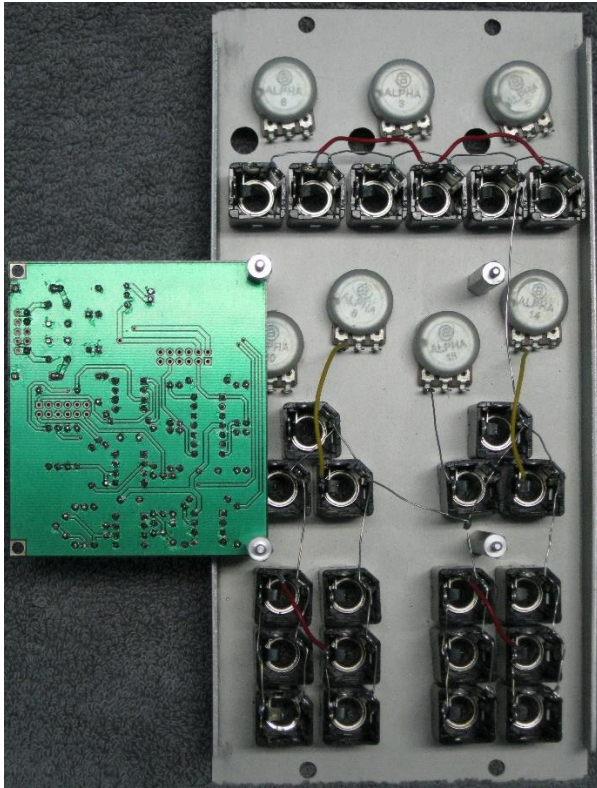
Note: My pattern of GND wiring is a bit different from the drawing.

Note: When using solid buss wire, always make contact with at least two sides of a terminal for a mechanical connection. Don't rely on just the solder.

Note: The extra length of wires from the Attenuverter wipers to the jacks allow the Secondary PCB to be added later.

Wiring:

- 1) Hang the Main PCB on the left standoffs, back of the PCB facing up.



- 2) Simply wire point A on the PCB to point A on the Front Panel, etc..

Leave 1-2" slack if you plan to add the LEDs later.

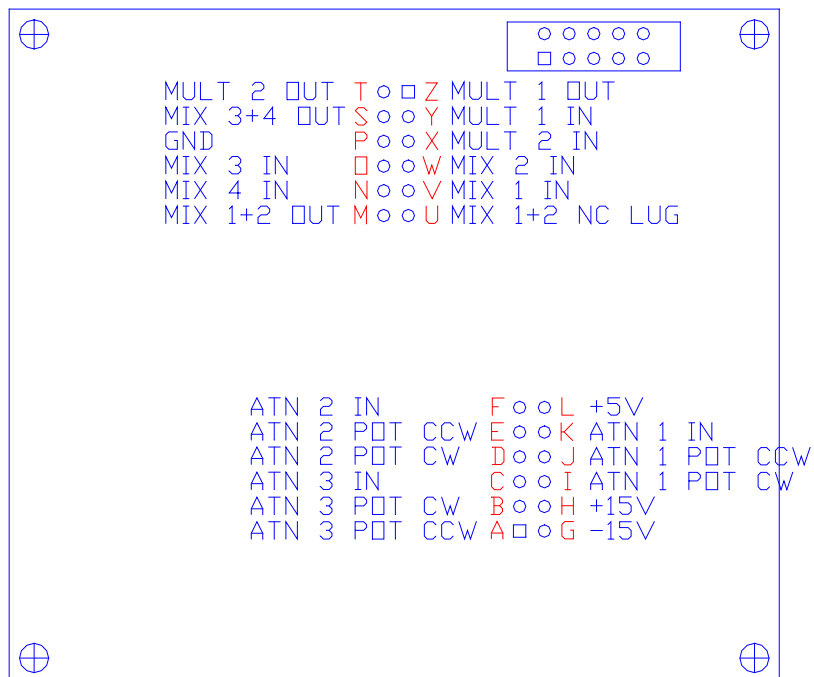
I use 24 AWG stranded wire for strength and flexibility.

To minimize numbers of colors to use, wire in group function, as in ATN2's three wires.

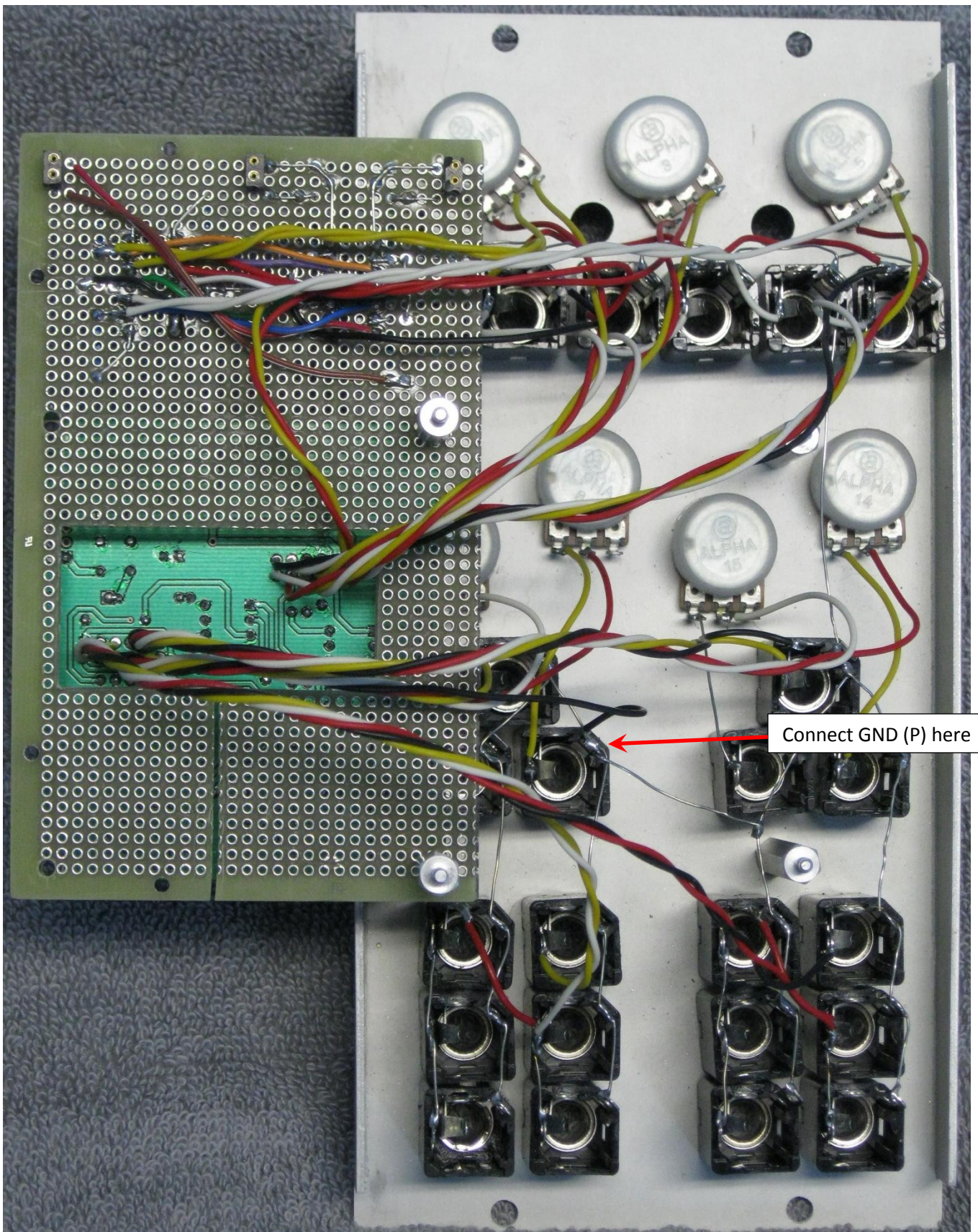
The groups I used were: ABC, DEF, IJKL and MUVW, NOPS, TXYZ.

H and G only run to the Secondary PCB.

BACK SIDE



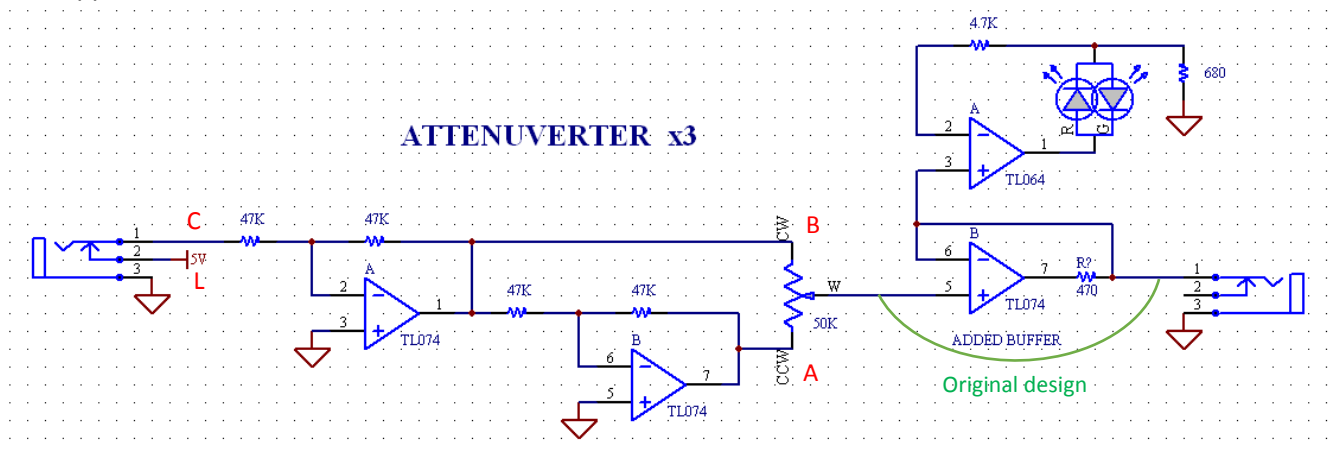
Main PCB Wiring Points (rotated for easier viewing)



Later...Secondary PCB added

Assembling the MVP with LEDs:

I chose to include the LEDs on the two kits I had and noticed that the Attenuverters' outputs weren't buffered so I added that (see schematic). Using Vector 8006 PCB material, a solid base to work on can be constructed that will support the LED drive, LEDs and the added Attenuverter Buffers.



Version #1 – Being a die-hard DotCom guy, for the first one used a perf board Secondary PCB that the Main PCB could plug its headers into and MTAs that the controls and jacks could plug into. It also provided a way to mount the LEDs and associated circuitry because the LEDs were too small for the panel. Aaaand it was a real pain to build.

Version #2 – For this version, I'll use the lessons learned and just hardwire the main PCB to the controls and jacks but still use a Secondary PCB to mount the extra circuitry and LEDs.

Extra Part Information (from Mouser):

Description	Number used	Mouser Number	Price ea (June 2017)
Standoff, M/F, 1.25"s	4	534-8407	\$0.87
Vector PCB	1	574-8006	\$47.02
Socket, IC, 14 pin	2	575-199314	\$0.46
Opamp, TL074	1	595-TL074BCN	0.862
Resistor, 1%, 1/4W, 470ohm	3	603-MFR-25FRF52-470R	0.12
Capacitor, 0.1uF, 50V	4	594-K104K15X7RF53L2	\$0.044

LED Choice:

The synthCube kit provides a small perf board on which to mount the LED drive opamp and resistors that are on the Eurorack version Control PCB. But there's no way to mount the PCB. The two kits I have kit provided an assortment of LEDs: two T1-3/4 Bi-Color LEDs, two T1 Bi-Color LEDs and three snap-in red LEDs; these last ones are the only ones that would fit the 0.312" (5/16") diameter holes in the panel. Since I wasn't sure what the T1-3/4 LEDs were, I got an assortment of LEDs and lenses to see what I liked best. Using Vector 8006 PCB material, a solid base to work on can be constructed that will support the LED drive, LEDs and the Attenuverter buffers.

I went with a curved lens, retainer clip, LED combo that cost about \$2 each. A bit difficult to mount and align but I like the look. Another choice could be a flat-faced, snap-in type that would be easier to mount but more difficult to wire that also costs about \$2 each. And there are other choices such as a domed lens or even an 8mm LED just remember the LED should be bi-colored.

LEDs #1 (I used this one)

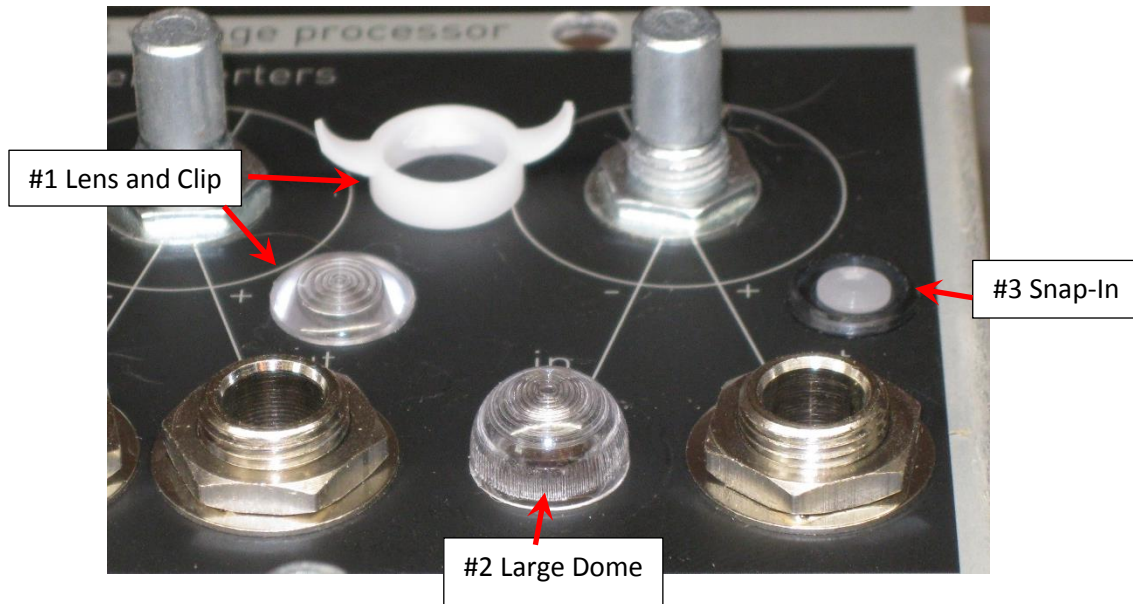
Description	Number used	Mouser Number	Price ea (June 2017)
LED, Bi- Color	3	604-WP7113SRSGW	\$0.47
LED Lens, Clear, Dif.	3	607-4742	\$0.77
Lens Retainer Clip	3	607-4740	\$0.73

LEDs #2 (requires drilling panel to 0.354-0.360")

Description	Number used	Mouser Number	Price ea (June 2017)
LED, Bi- Color	3	604-WP7113SRSGW	\$0.47
LED Lens, Snap-In, Clear, Dif.	3	607-4342	\$0.96

LEDs #3

Description	Number used	Mouser Number	Price ea (June 2017)
LED, Bi Color, Snap-In	3	696-SSI-LXH387HGW	\$1.92



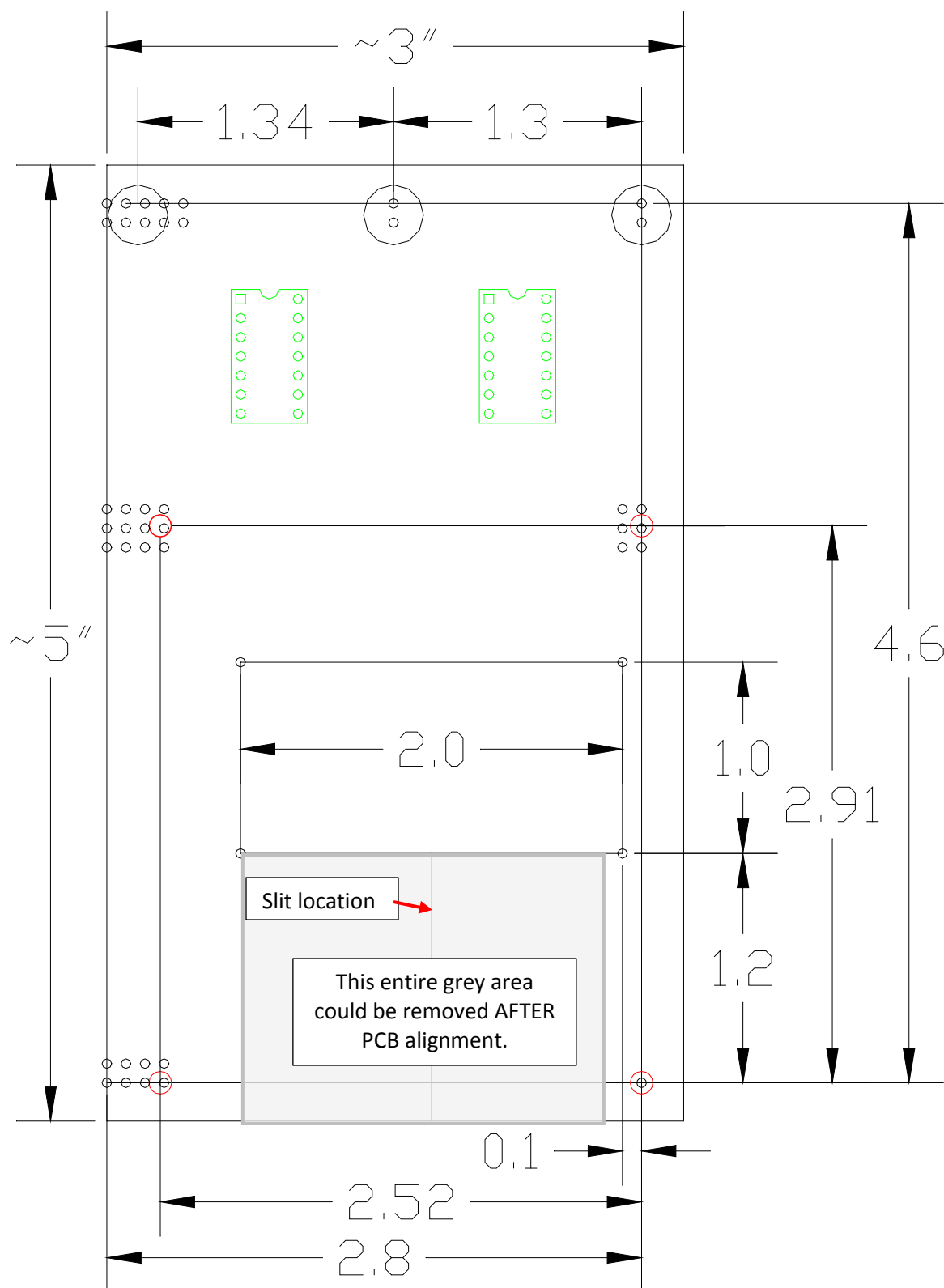
Secondary PCB Preparation:

It may seem like a waste of material but the size of the secondary PCB provides the stability for the LEDs (if you use the one I selected). But this gets in the way of the wiring. To address this, cut a 2"x 1.5" hole out of the center of the secondary PCB as shown. File the edges smooth.

(Original Note): Once the PCB fits properly on the standoffs you can cut a slit so the wires will pass through the hole. Then the slit may be "stitched" back together for strength.)

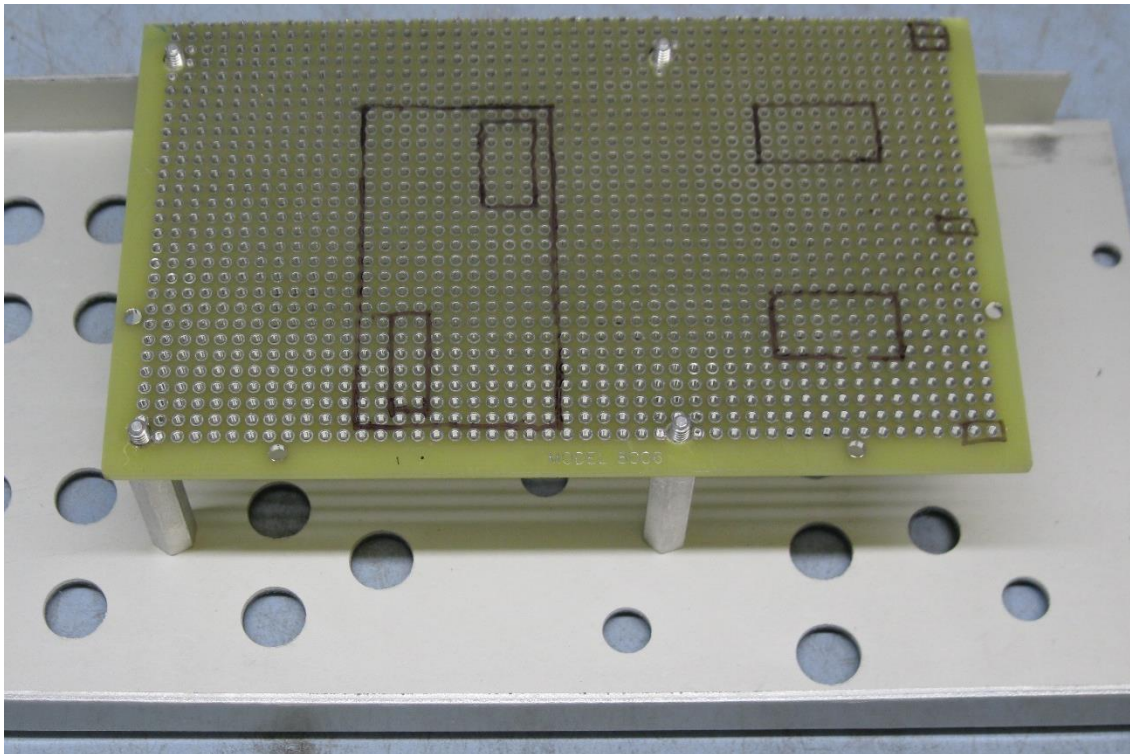
(Updated Note): It turned out that there's enough strength in the PCB that a larger chunk may be removed (and saved for later). See the drawing below.

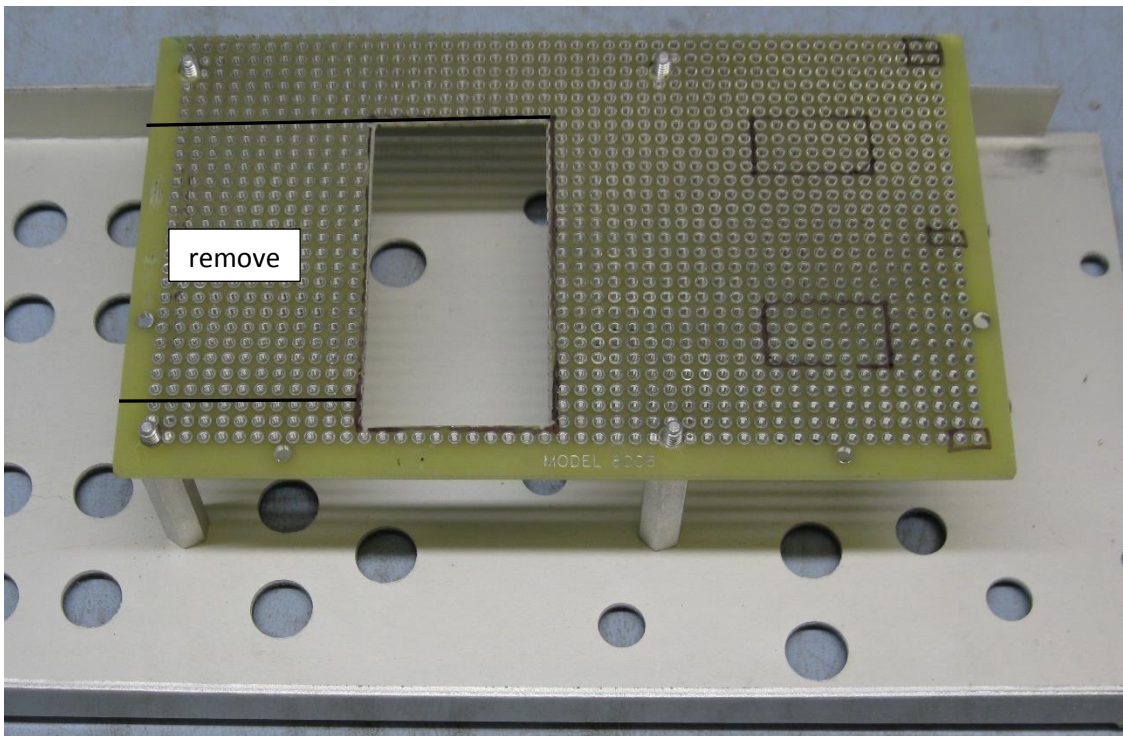
Use the Main PCB as a template (the power connector goes in the upper right corner), the template you created earlier or the drawing below to mark and drill the perf board. I used the end of a Vector 8006 such that the length of the side to side holes are 2.8" leaving bare PCB material around the top, right and bottom sides.



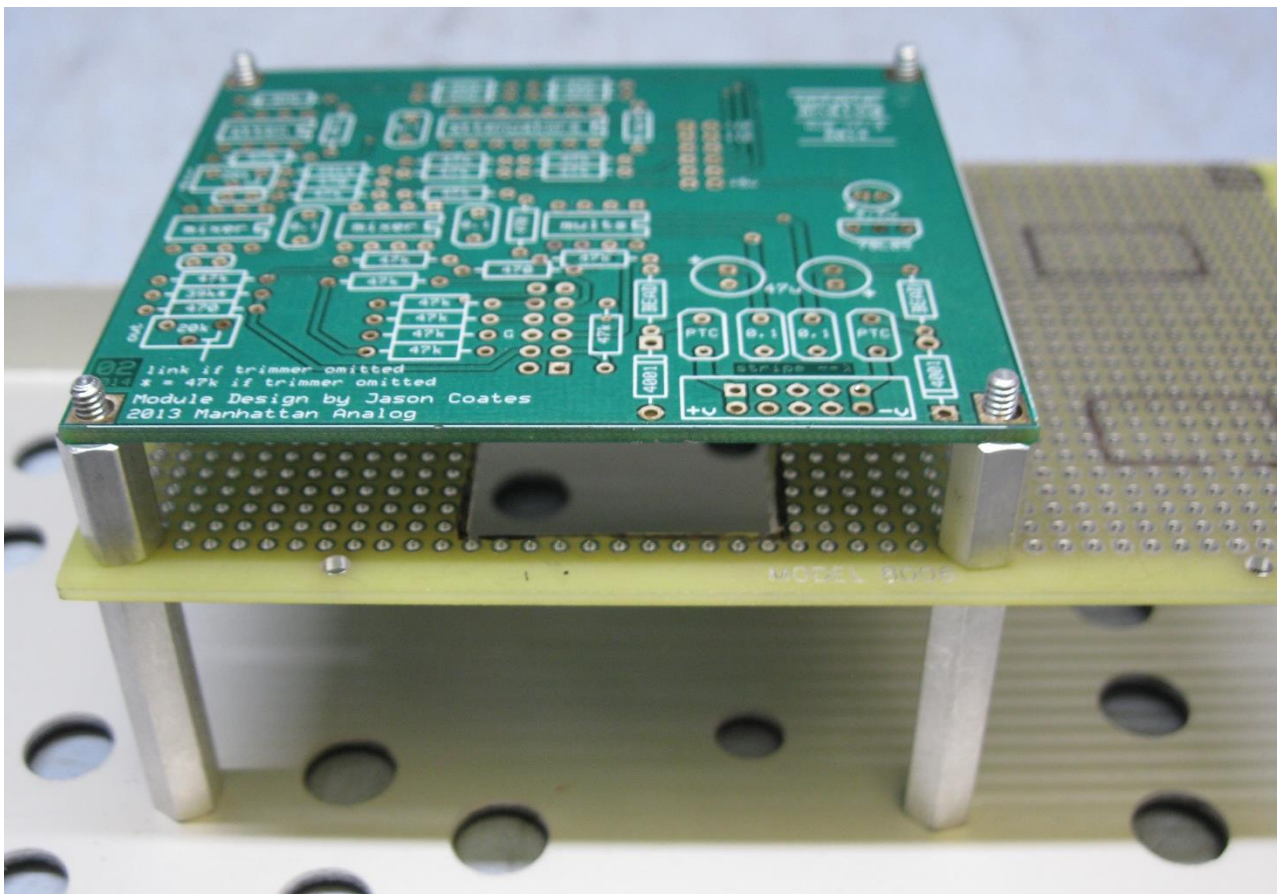
Secondary PCB (in inches, not to scale)

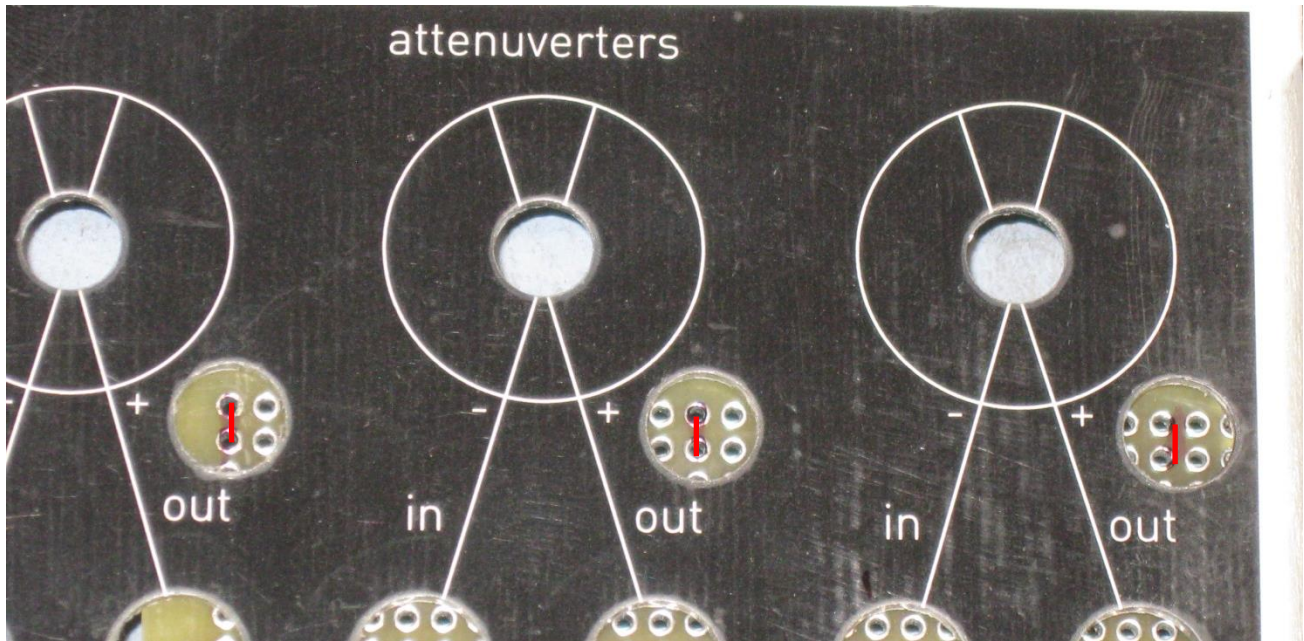
- 1) Measure, score and snap the PCB then use a file to clean up the edge.
- 2) Align the Main PCB Lower Right mounting hole with the lowest, rightmost via in the 8006 material.
- 3) Mark the center of the other three mounting holes. Note that the Lower Left and Upper Right holes are off from the perf board holes by about 0.020" in either the X or Y direction. And the Upper Left hole will be off diagonally by 0.020" in both X and Y.
- 4) Drill out the four closest vias to where you marked the mounting holes using a 7/64" bit. Using the Lower Right hole as the key alignment hole allows a very tight alignment. You may drill out with a 1/8" bit later for a looser fit.
- 5) Start with the Lower Left and Upper Right holes and, using a sharp knife (X-Acto) or gentle drilling, shape them into ovals that align to the main pcb holes. Tip: Install three standoffs in the front panel to help with the alignment.
- 6) Then work on the Upper Left until it also aligns.





- 11) Mark the PCB for the center cutout as shown.
- 12) Use a Dremel with a cutoff wheel, drill and nibbler or some other means to create the hole.
- 13) Smooth the edges with a file. This allows the wires from the Main PCB to pass through the Secondary PCB.

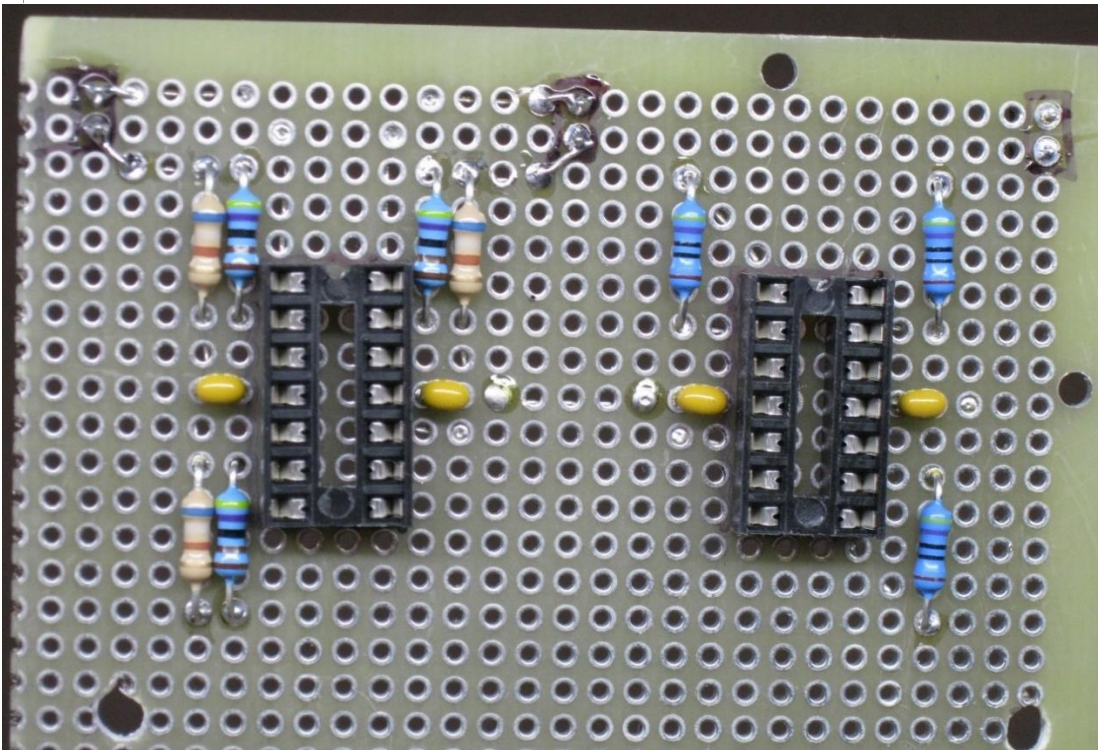
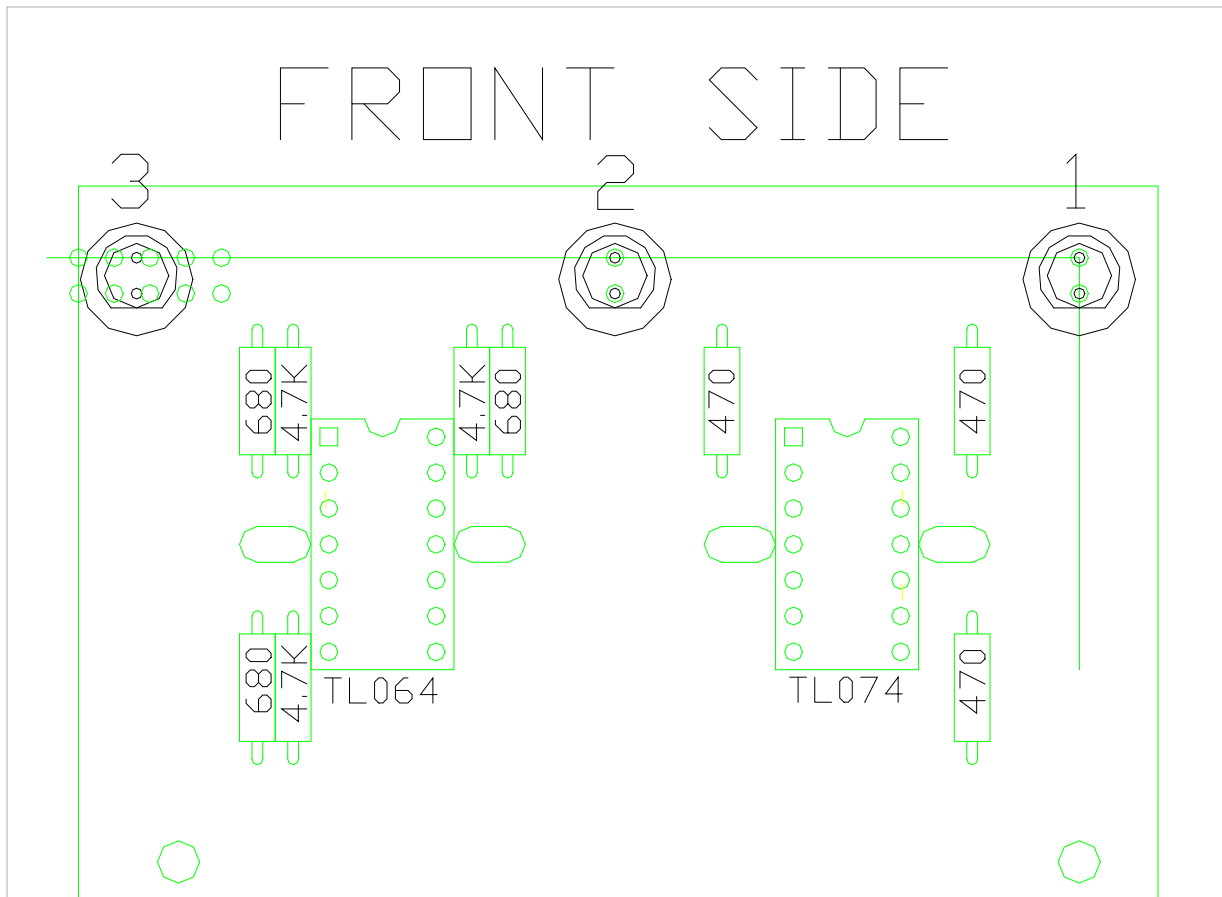




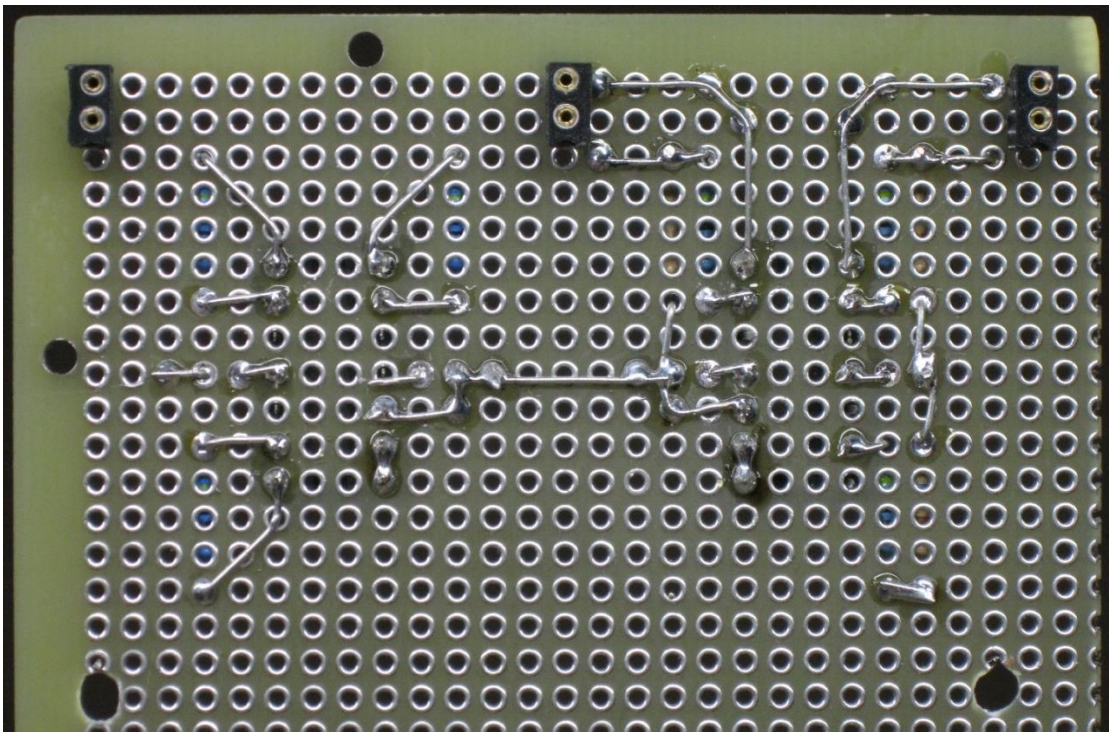
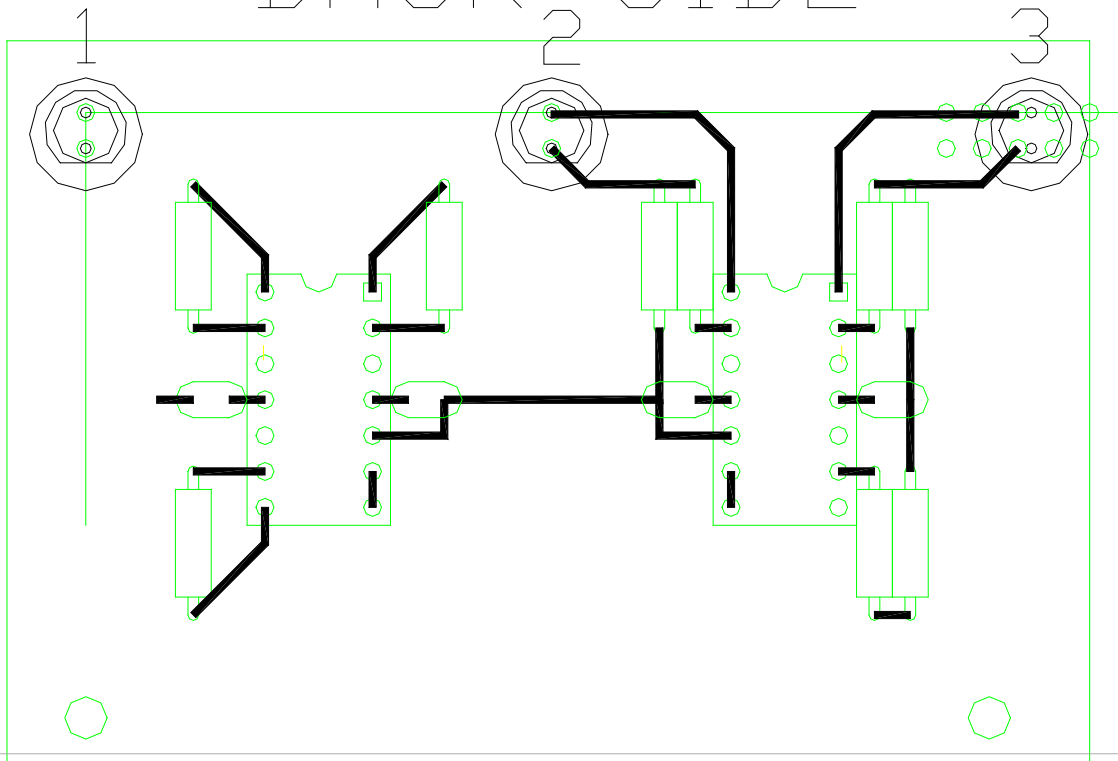
- 8) Remove any standoffs you may have used. Use four #4 screws top align the Secondary PCB directly to the front panel.
- 9) From the front of the panel, mark the center of the LED holes. (On my panels the center-to-right hole distance was about 0.030-0.040" wider than the left-to-center hole distance.) The left and center holes should line up with a vertical column of vias while the right may be in-between the columns. They may also be off about 0.010" vertically.
- 10) You may now enlarge the mounting holes if you wish.
- 11) If you have built the No LED version so far, slit/cut the PCB where shown. This allows you to slide the Secondary PCB over the wires.

Building the Secondary PCB:

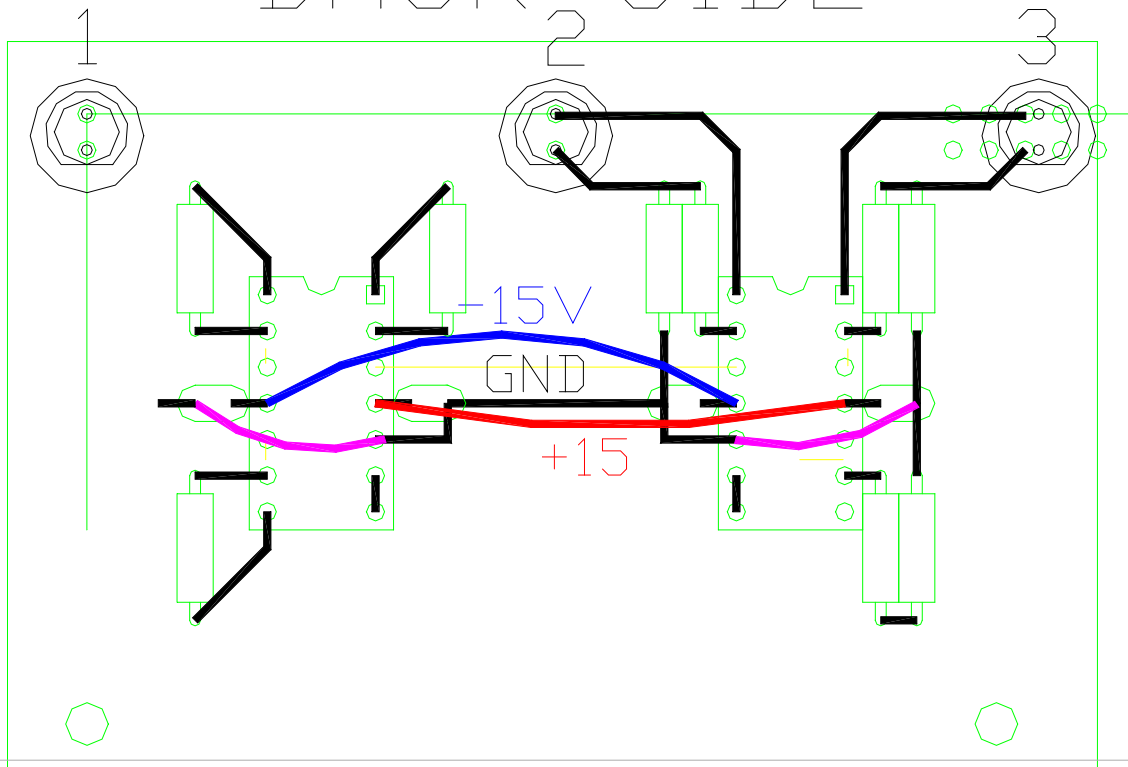
- 1) Install the two IC Sockets (from the list above) as shown.
- 2) Break the machined pin IC socket included in the kit into three 2 pin sockets for the LEDs.
- 3) Install the LED sockets on the rear of the PCB where you previously marked them (to the closest set of holes).
- 4) Install the resistors and caps where shown. Use the component leads to make the easy connections to the IC sockets and LEDs.
- 5) Use wire (I use multi-color ribbon cable wire) to connect the rest of the connections on the PCB.



BACK SIDE

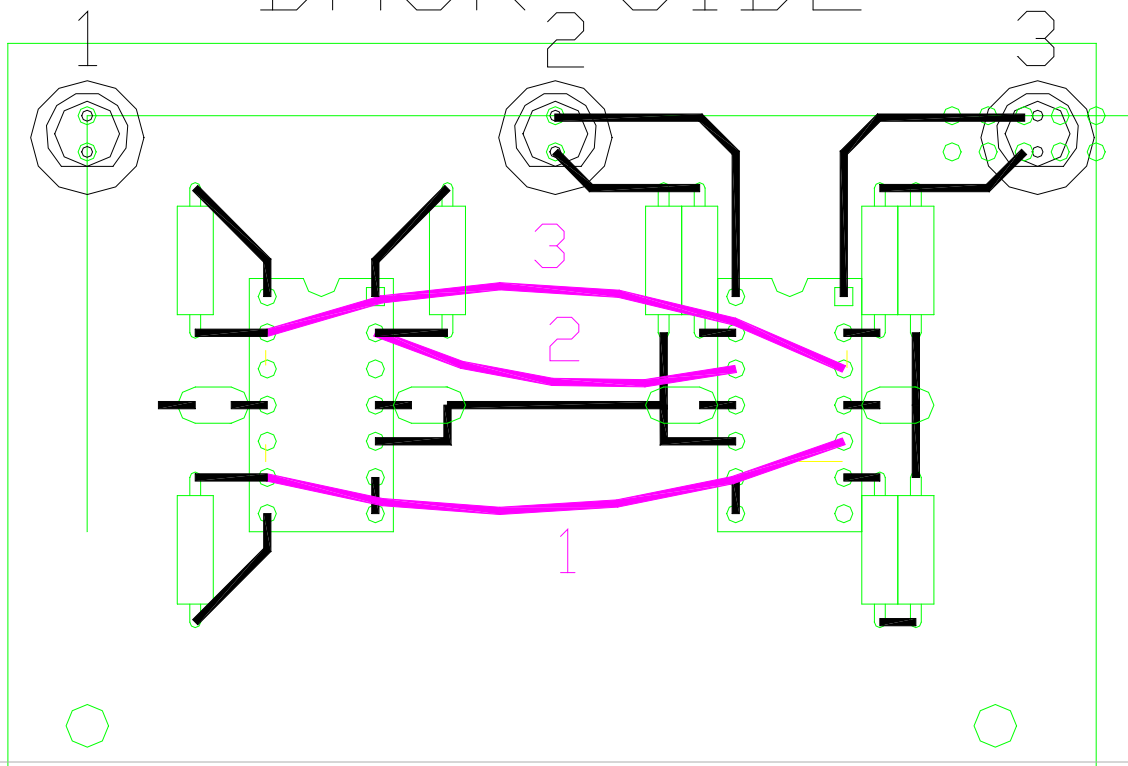


BACK SIDE

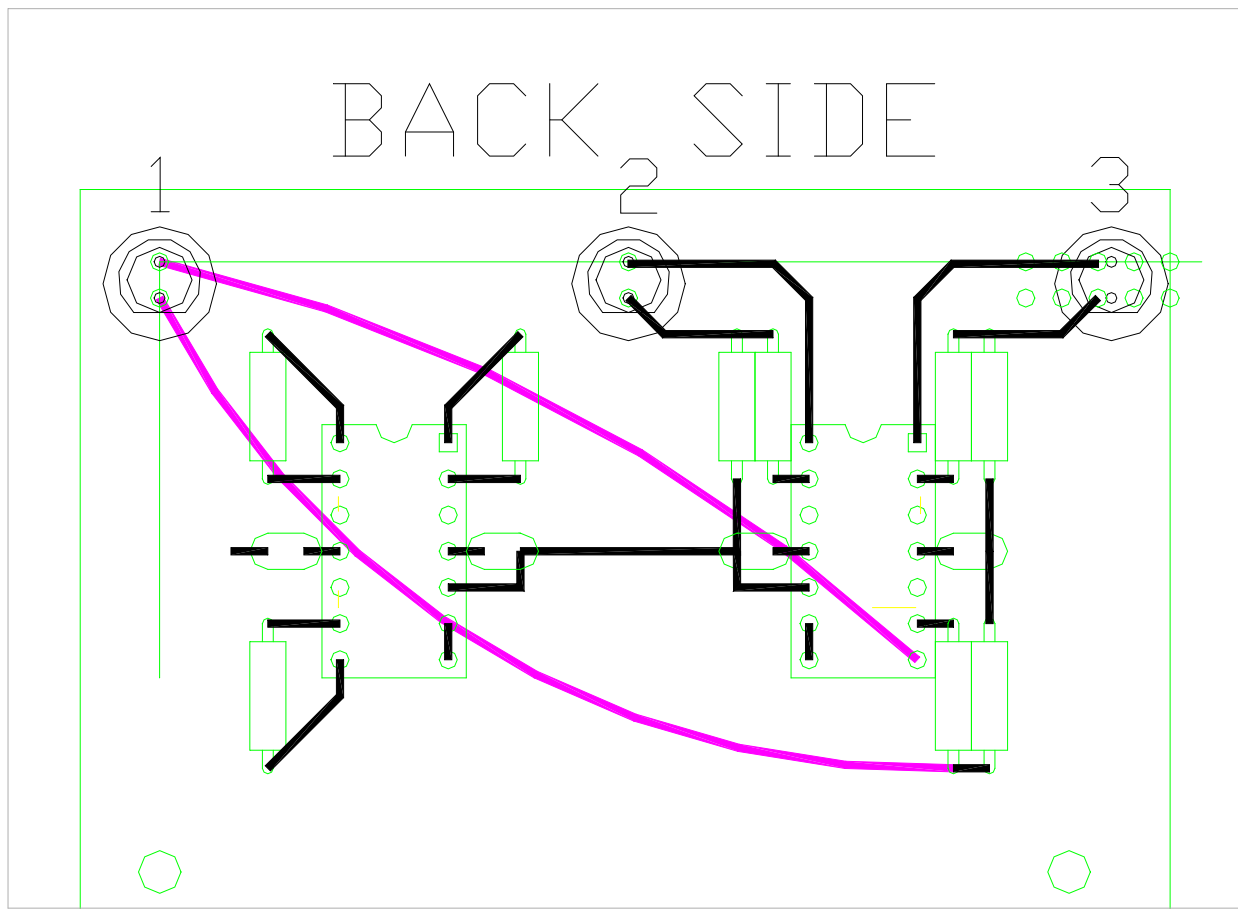


Power Jumpers

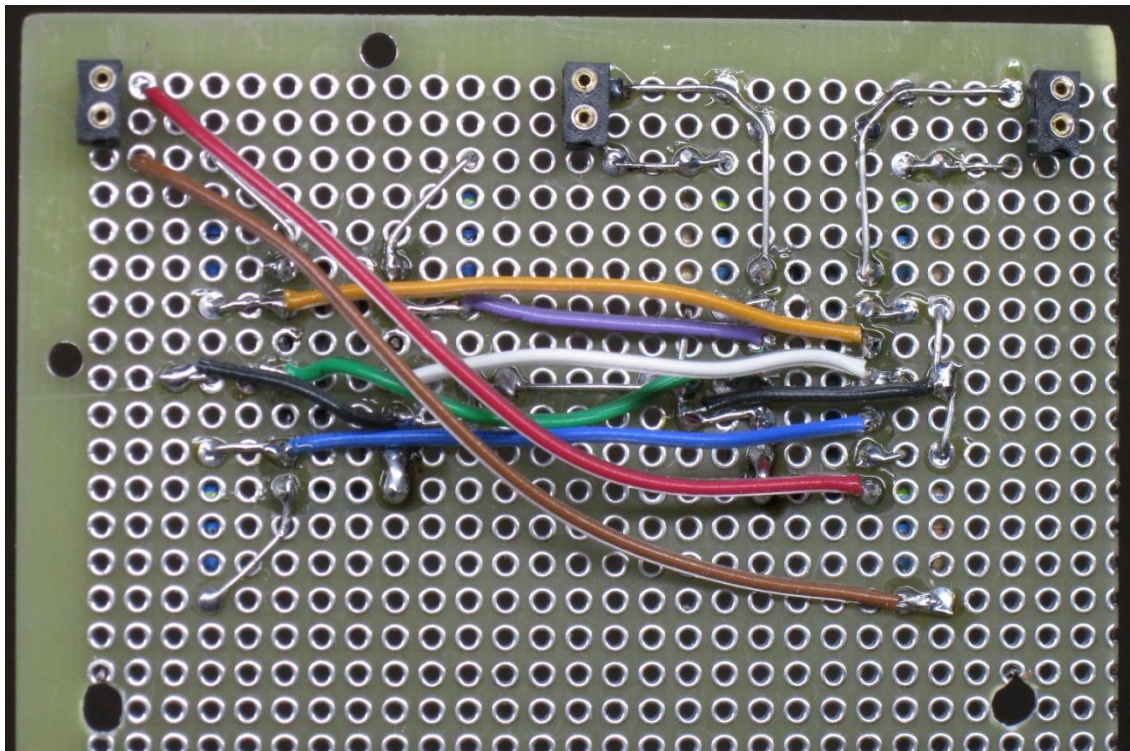
BACK SIDE



LED Drive Input Jumpers



LED 1 Wires



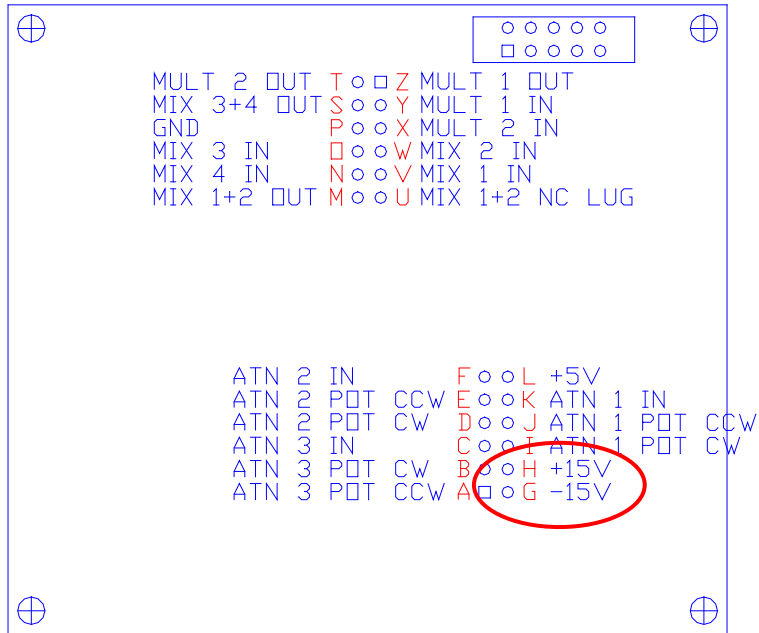
Completed Secondary PCB

Wiring in the Secondary PCB

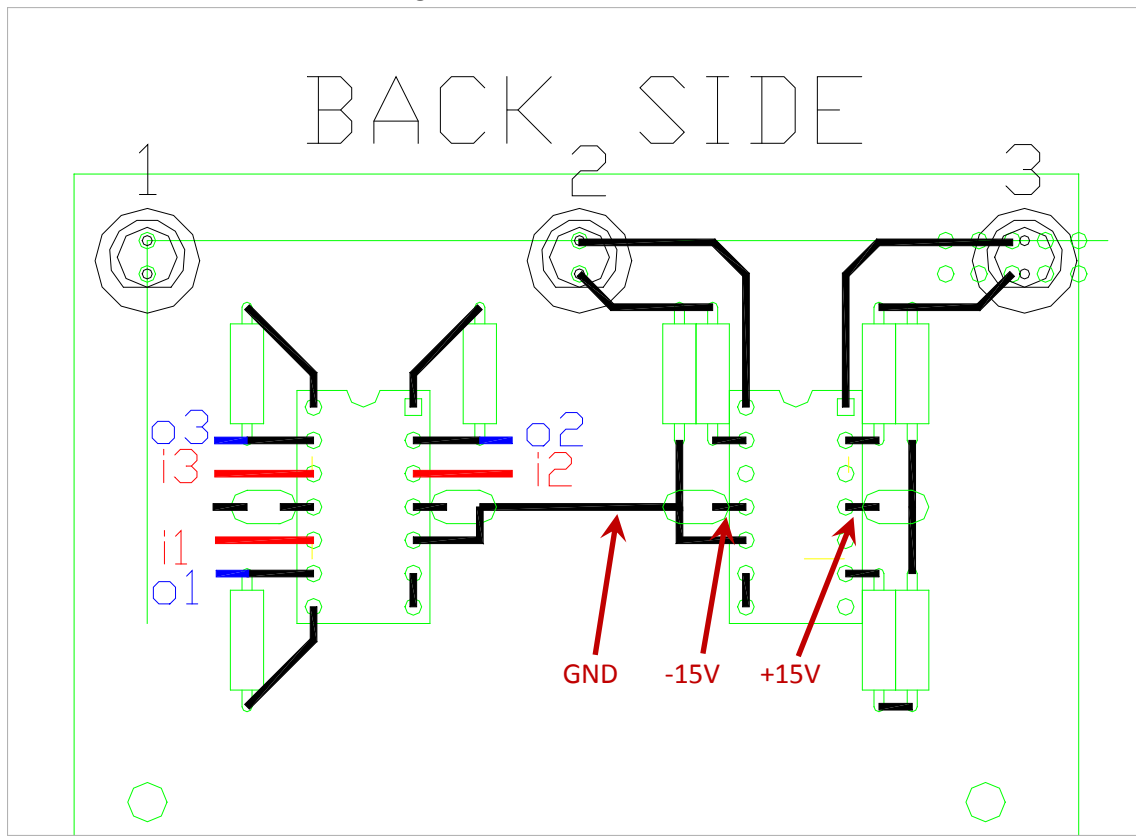
You've already wired the Main PCB to the Front Panel, you just need to add three wires.

- 1) From the Main PCB, attach wires to points H (+15V) and G (-15V)
- 2) On the Front Panel, attach a wire to the Ground wire daisy chain around point P.
- 3) Attach those three wires to the proper locations on the Back Side of the Secondary PCB.
- 4) Cut the Attenuverter wiper wires that you left long earlier at their middles and wire them to the proper locations on the Back Side of the Secondary PCB.

BACK SIDE



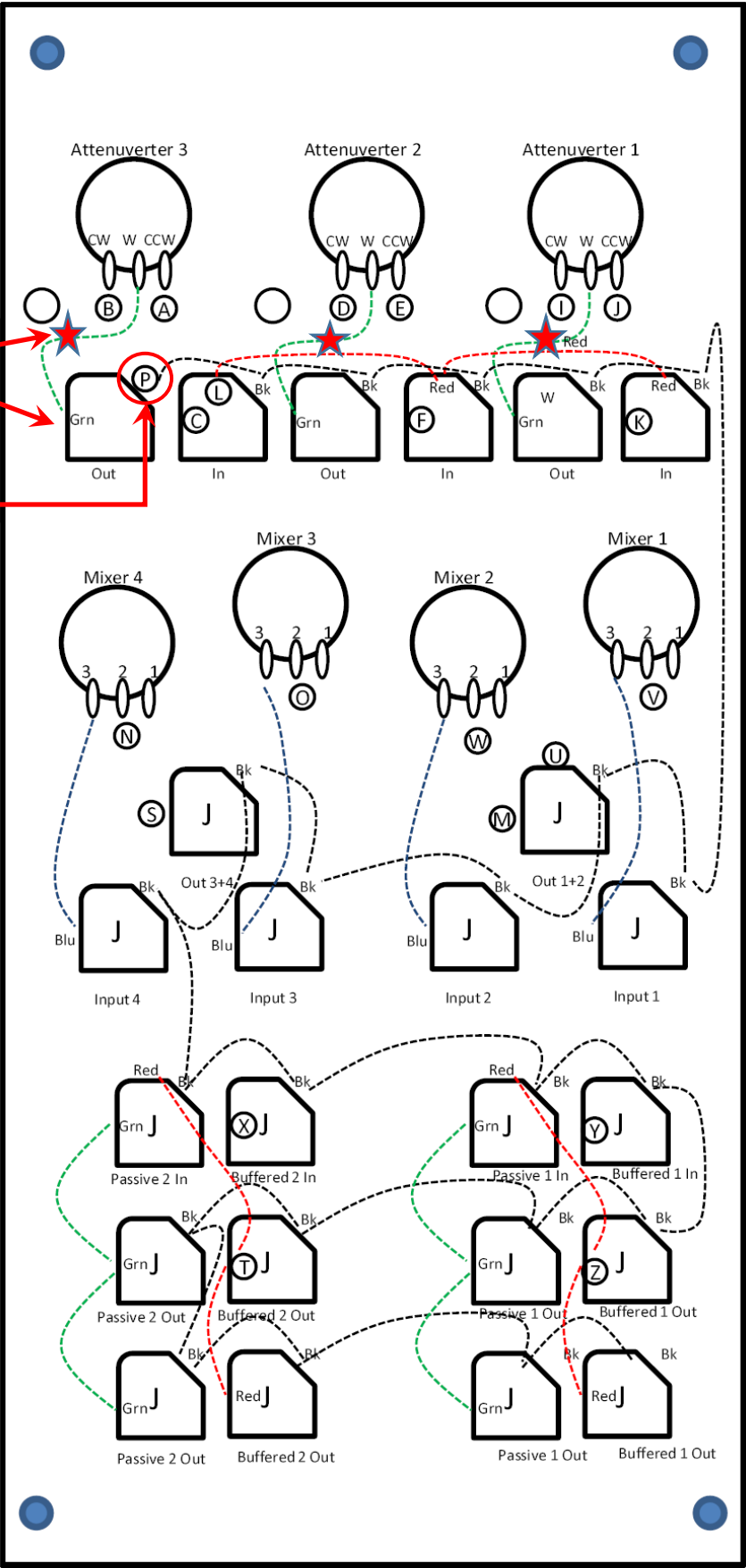
Main PCB Wiring Points



Buffer Input/Output and Power Connections

Wiper goes to Buffer on Secondary PCB.
 Output of Buffer connects to Output Jack.
 Ground connects here.

Image provided by
 synthCube



MA MVP Rear View of Panel- dotcom/MU

Installing the LEDs:

The LED leads need to be trimmed to length. Do this one LED at a time for the best fit.

This method describes the LED/Lens combination I selected; if you made a different choice, the method would be similar from Step 3 on.

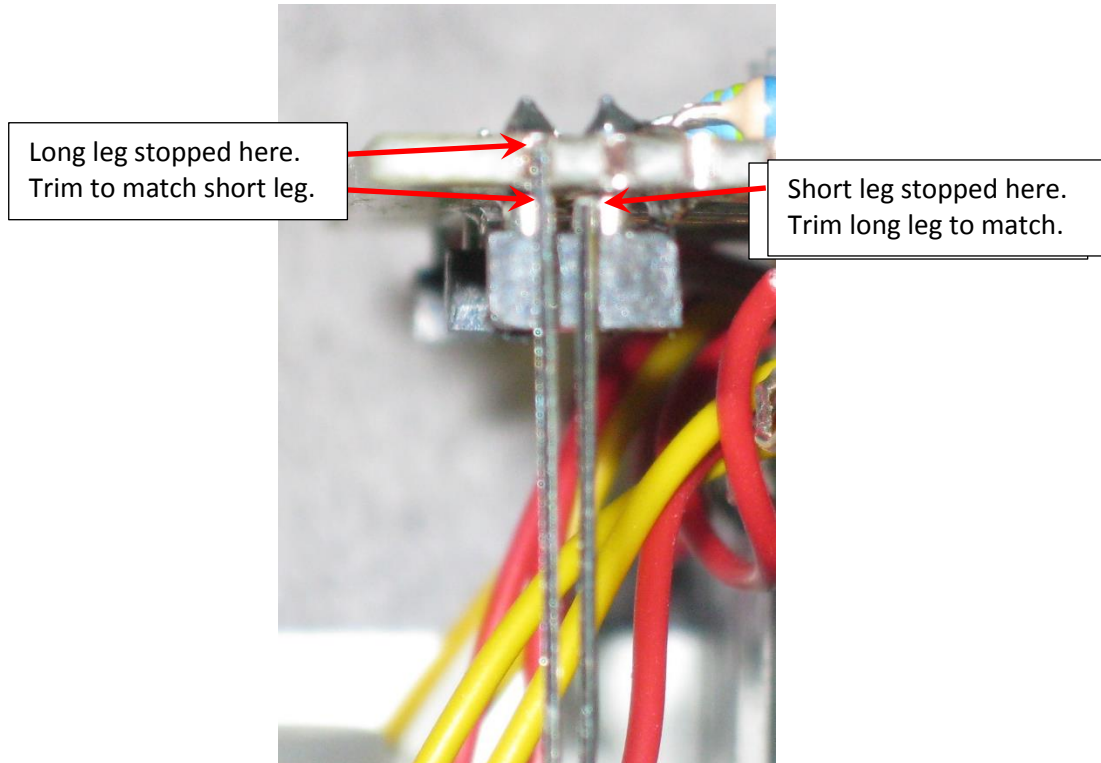
1) Mount the Lens and Clip into the Front Panel. You can add a drop of epoxy to prevent the clip from popping off.

2) Place an LED into the Lens (flatted edge of the LED (short leg) goes towards the bottom of the panel).

3) Place the Secondary PCB into position on the 1.25" standoffs with the LED leads to one side of the LED socket.

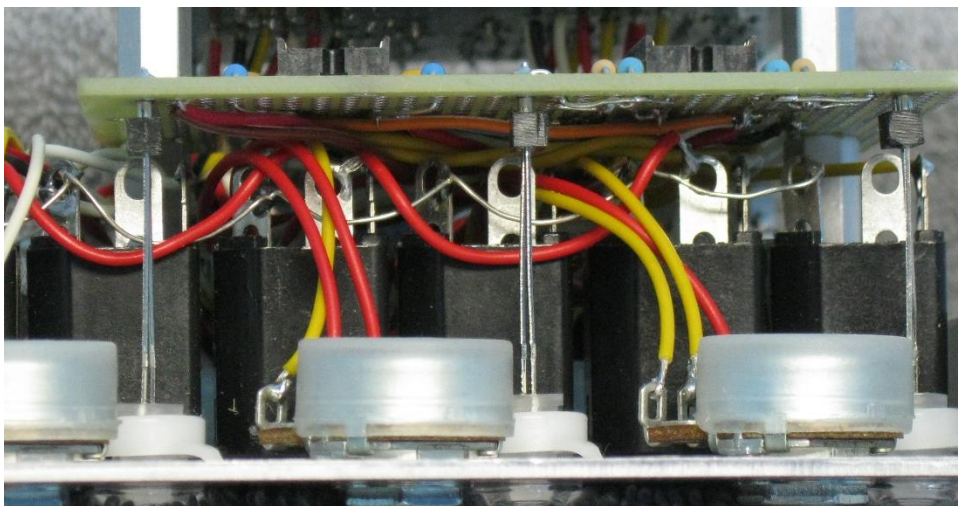
The LED leads will begin to make contact with the socket about halfway into the socket plastic part and will bottom out almost at the bottom of the PCB.

4) Making sure the LED is fully seated in the Lens, trim the LED leads at about halfway between the bottom of the socket plastic and the PCB.



5) Test fit the LED into the socket and check that it fits into the Lens properly.

6) Repeat for the other LEDs.

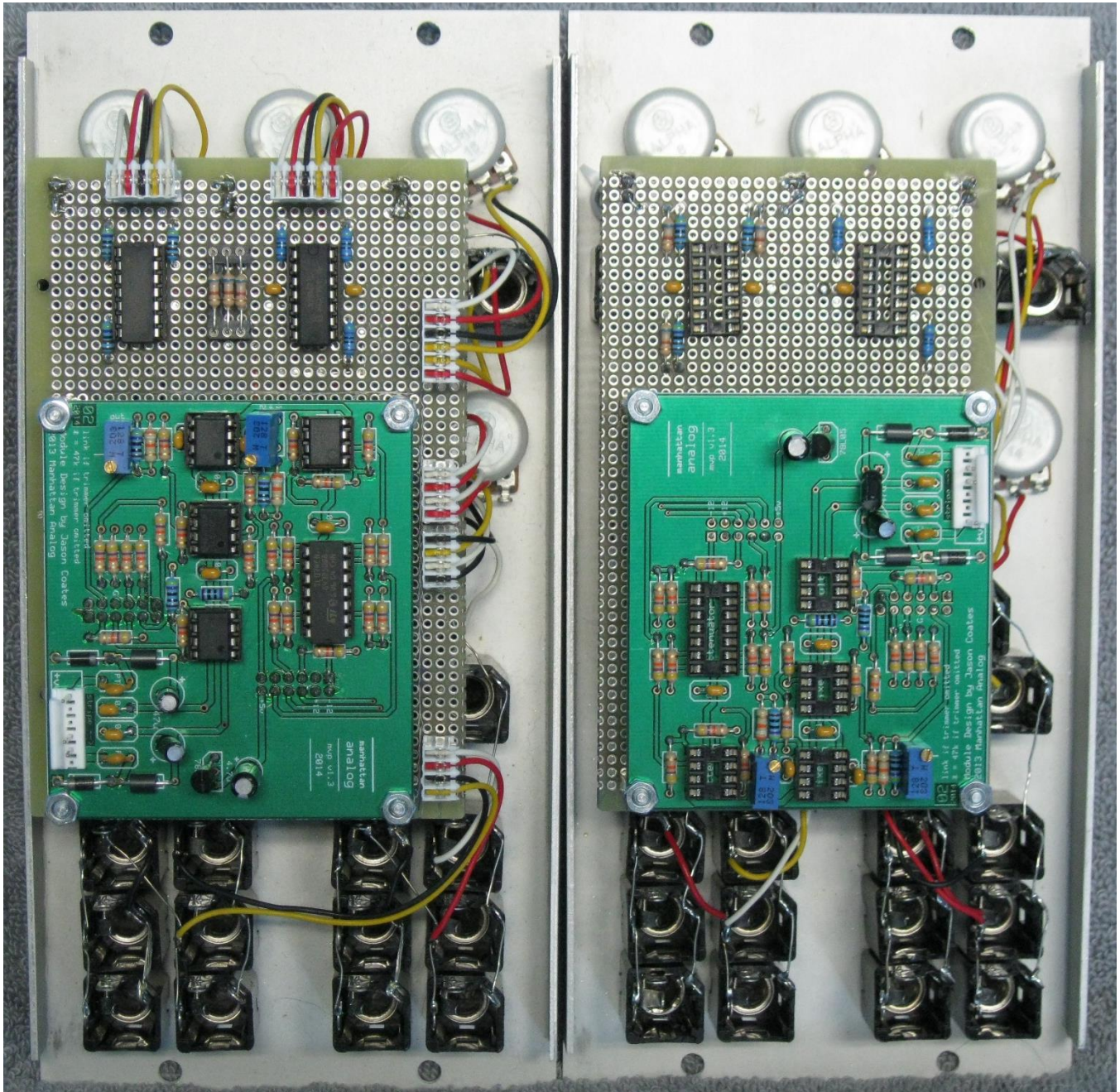


Calibration:

The two trim pots only affect the Mixer section. The gain range is about 85% to 190% of a single input with the input level pot set fully clock-wise. There's a note on the PCB that puts a jumper in place of the trim pot and changes the 39K to a 47K which would provide unity gain. There's a lot of possible variations you could do but I'll keep it simple here by setting unity gain and you can experiment on your own.

- 1) Connect a voltage source (about 5V) or signal to the Mixer 1 input.
- 2) Monitor Mixer 1+2 Output and adjust the trimpot in the middle of the board to achieve the same level.
- 3) Monitor Mixer 3+4 Output and adjust the trimpot at the bottom of the board to achieve the same level.

These are how mine look:



1st version with MTAs

This version with hardwires